

Automatic Disturbances Rejection Controller for Precise Motion Control of Permanent-Magnet Synchronous Motors

Y. X. Su, C. H. Zheng, and B. Y. Duan

Abstract—A highly robust automatic disturbances rejection controller (ADRC) is developed to implement high-precision motion control of permanent-magnet synchronous motors. The proposed ADRC consists of a tracking differentiator (TD) in the feedforward path, an extended state observer (ESO), and a nonlinear proportional derivative control in the feedback path. The TD solves the difficulties posed by low-order reference trajectories which are quantized at the sensor resolution, and the ESO provides the estimate of the unmeasured system's state and the real action of the unknown disturbances only based on a measurement output of the system. Simulations and experimental results show that the proposed ADRC achieves a better position response and is robust to parameter variation and load disturbance. Furthermore, the ADRC is designed directly in discrete time with a simple structure and fast computation, which make it widely applicable to all other types of drives.

Index Terms—Disturbance rejection, friction compensation, nonlinear proportional derivative (NPD) control, observers, permanent-magnet synchronous (PMAC) motor.

I. INTRODUCTION

WITH the rapid progress in power electronics, microprocessors, and control theory, permanent-magnet synchronous (PMAC) motors, which possess the characteristics of high power/weight ratio and inherent maintenance-free capability, have been recognized as one of the key components in automation and robots [1]–[9]. Conventionally, the linear control schemes dominate the controller market for PMAC motors [2]. However, the linear control design is based on a linear model and on the assumption that the mechanical and the electrical time constants differ at least for one order of magnitude. Apart from the fact that its formal treatment is inconsistent, linear control also does not allow a methodical design of the control structure and the determination of the optimal control parameters. The influence of nonlinearities, external disturbances (mechanical torque), and parameter and

structural perturbations (inertia, friction) are usually considered only by the demanded “adequate robustness” of linear controllers. Consequently, the linear control cannot assure a satisfactory dynamic behavior in the entire operating range [2], [7]. Recently, many researchers have contributed their efforts toward high-precision control of PMAC motors, and various algorithms have been proposed [1]–[9].

In recent years, due to the great advances in nonlinear control theory, the observer-based controller has become one of the most commonly used schemes in industrial applications. The disturbances observers offer several attractive features. In the absence of large model errors, they allow independent tuning of disturbances rejection characteristics and command-following characteristics. Further, compared to integral action, disturbances observers allow more flexibility via the selection of the order, relative degree, and bandwidth of low-pass filtering [10]. Due to these attractive features, the observer-based controller has become one of the most commonly used schemes in control systems and an enormous amount of results on them have been published [9]–[17].

Furthermore, unmodeled nonlinear friction has been found to have an unwanted influence on control performance [2], [11], [18]–[21]. Usually, a model-based approach is employed to reduce the effect from friction [2], [18], [19]. Even in adaptive control of servo systems, an initial friction model is also crucial to ensure smooth control signals and rapid parameter convergence. However, in practical conditions, because of the inherent nonlinearity of friction, precise friction modeling has always been a difficult and challenging problem. Recently, extensive efforts have been made toward more advanced control of precision motion systems, based on a disturbances observer to implement the friction compensation [11], [20], [21].

In this paper, a discrete-time automatic disturbances rejection controller (ADRC) is designed to realize high precision motion control of PMAC motors. The nonlinear friction is considered as an unknown disturbance to the system, and an extended state observer (ESO) in the feedback path provides the estimate of the unmeasured state and the real action of the unknown nonlinear friction, only based on a position measurement. A tracking differentiator (TD) in the feedforward path solves the difficulties posed by low-order reference trajectories which are quantized at the sensor resolution, and a nonlinear proportional derivative (NPD) control is used to synthesize the control action. The effectiveness of this control scheme is verified by simulations and experimental results performed on a PMAC motor.

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A. TD Design

In many practical situations, the performance of the controlled system is limited by how to select the high-quality differential signal from the noncontinuous noisy measured signal. In general, the differential signal is usually obtained by the backward difference (BD) of the given signal, which is very noisy and limits the overall performance [20]–[29]. To limit the amount of this noise, it is necessary to filter the differential signal. Additional delay is an unavoidable drawback of these conventional filters used frequently in industrial electronics [23]–[25]. This delay is particularly harmful when the filtered primary signal is to be used for time-critical feedback or synchronization purposes.

In this section, a TD [26] is used to provide a high-quality differential signal for more effective and robust performance in the presence of measurement noise. Although a rigorous stability analysis of TD has not been available, the viability for practical applications has been demonstrated via simulations and experiments [21], [22], [26]–[28].

For a boundary integrable function $r(t)$, system (9) can be used as a high-performance TD to provide two signals $r_1(t)$ and $r_2(t)$, such that $r_1(t) \rightarrow r(t)$ and $r_2(t) \rightarrow \dot{r}(t)$, respectively [26]–[28]

$$\begin{cases} r_1(k+1) = r_1(k) + hr_2(k) \\ r_2(k+1) = r_2(k) + hfst(r_1(k) - r(k), r_2(k), \delta_0, h_0) \end{cases} \quad (9)$$

where h is the sampling step, k denotes the k th sampling instant, h_0 a filtering factor which determines the suppression of noise, and δ_0 a velocity factor to determine the transition characteristics. The function $fst(x_1, x_2, \delta_0, h_0)$ is defined as

$$fst(x_1, x_2, \delta_0, h_0) = - \begin{cases} \delta_0 \operatorname{sgn}(a), & |a| > d \\ \delta_0 \frac{a}{d}, & |a| \leq d. \end{cases} \quad (10)$$

in which, $\operatorname{sgn}(\cdot)$ denotes a standard sign function, and a and d can be determined as follows:

$$a = \begin{cases} x_2 + \frac{(a_0 - d)}{2} \operatorname{sgn}(y), & |z| > d_0 \\ x_2 + \frac{z}{h_0}, & |z| \leq d_0. \end{cases} \quad (11)$$

with

$$\begin{cases} d = \delta_0 h_0 \\ d_0 = h_0 d \\ z = x_1 + h_0 x_2 \\ a_0 = \sqrt{d^2 + 8\delta_0 |z|}. \end{cases} \quad (12)$$

It is noted that the developed TD has high robustness to the variation of the design parameters δ_0 and h_0 . In general, large δ_0 is helpful to speed the transition and tracking, and large h_0 helps with the noise suppression. δ_0 can be chosen from the range of [5 10 000], and $h_0 = (2\text{--}25)h$. There is a tradeoff between the tracking and filtering.

The performance of the TD is shown in Fig. 3. The reference input is $r(t) = \sin(2t)\text{rad}$, $t = kh$, and it is perturbed by an additive white-noise component with the maximum amplitude of 0.01 rad. For comparison, the differential signal obtained by the

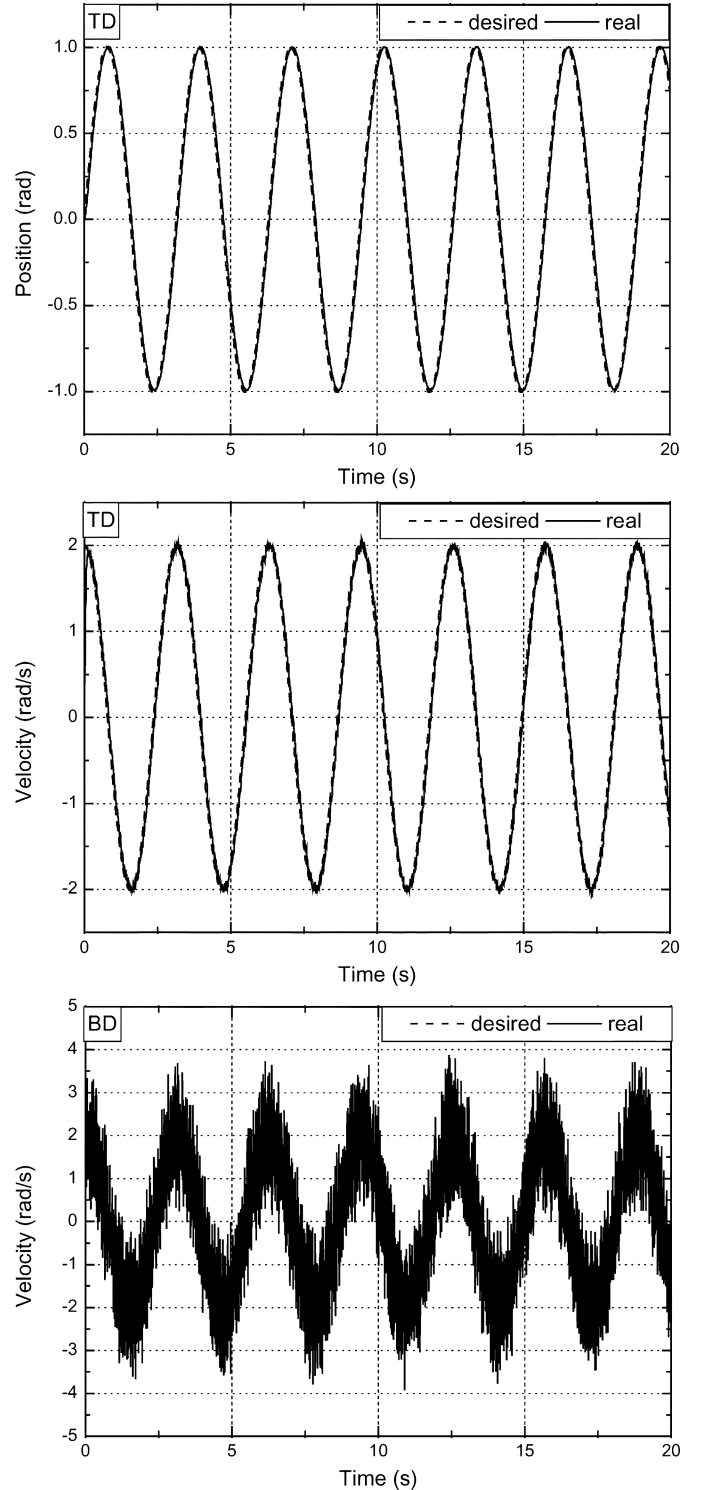


Fig. 3. Comparison of TD with general BD.

general BD method is also plotted in this figure. The simulations were programmed in Matlab with a fourth-order Runge–Kutta and run on a PC Celeron-433. The simulation step was determined as $h = 0.005$, and all the initial values of r_1 and r_2 were set to zero. The design parameters for the TD were determined as follows: $h_0 = 5h$ and $\delta_0 = 6000$. It can be seen that the differential signal obtained by the developed TD is much better compared to that of the conventional BD method.

B. ESO Design

Consider a discrete uncertain nonlinear system with unknown disturbance

$$y^{(n)} = f(y, \dot{y}, \dots, y^{(n-1)}, k) + d(k) + b_0 u(k) \quad (13)$$

where $f(y, \dot{y}, \dots, y^{(n-1)}, k)$ is an unknown function, $d(k)$ is an unknown disturbance, $u(k)$ is the input to the system, and b_0 is a system parameter. Augmenting the variable $y^{(n)}$ into the state of the system, then the augmented state is the serial derivative of the measured state y . As a result, the $(n+1)$ th-order ESO for this system based on the measured output $y(k)$ only can be constructed [28], [29], as shown by (14), at the bottom of the page, where $\alpha_i > 0 (i = 1, 2, \dots, n)$, $\beta_j > 0 (j = 1, 2, \dots, n+1)$, and $\delta_1 > 0$ are design parameters. The nonlinear function $\text{fal}(x, \alpha, \delta)$ can be defined as

$$\text{fal}(x, \alpha, \delta) = \begin{cases} |x|^\alpha \text{sgn}(x), & |x| > \delta \\ \delta^{(\alpha-1)} x, & |x| \leq \delta. \end{cases} \quad (15)$$

Note that, for this nonlinear ESO, it is an important fact that the extended state tracking the real action of the unknown disturbance, it is very helpful to realize the disturbance compensation. It is also noted that the developed ESO has high robustness to the variation of the design parameters. In general, $0 < \alpha_1 \leq 1$ and $\alpha_i = \alpha_1 / 2^{i-1} (i = 2, \dots, n)$, δ_1 can be chosen from the range of $[0.0001 \ 1]$, and small δ_1 is helpful for the noise suppression, $0 < \beta_j (j = 1, 2, \dots, n+1)$ can be determined based on the tracking performance, and large β_j is helpful to speed the transition and the tracking. There is a tradeoff between δ_1 and β_j for a better noise suppression and tracking performance, from a noisy disturbed measurement.

To validate the effectiveness of the ESO, a simulation on a disturbed nonlinear system is carried out. The disturbed nonlinear system can be described as

$$\ddot{y} = -y^3 - y - 0.2\dot{y} + d \quad (16)$$

where d is an unknown disturbance, which can be expressed as $d = 0.5 \text{sgn}(\cos(kh/2))$. For this second-order nonlinear system, the third-order ESO can be constructed as

$$\begin{cases} \varepsilon_0 = z_1(k) - y(k) \\ z_1(k+1) = z_1(k) + h(z_2(k) - \beta_1 \varepsilon_0) \\ z_2(k+1) = z_2(k) + h(z_3(k) - \beta_2 \text{fal}(\varepsilon_0, \alpha_1, \delta_1)) \\ z_3(k+1) = z_3(k) - h\beta_3 \text{fal}(\varepsilon_0, \alpha_2, \delta_1). \end{cases} \quad (17)$$

All the initial values of z_1 , z_2 , and z_3 were set to zero. For this third-order ESO, one feasible choice of the parameters was as follows: $\alpha_1 = 0.5, \alpha_2 = 0.25, \delta_1 = 0.00025, \beta_1 = 100, \beta_2 = 60, \beta_3 = 100$. The simulations were performed with and without the noise. The outputs z_1 , z_2 , and z_3 of the ESO and the outputs y_1 and y_2 of the system are shown in Figs. 4 and 5. It can be seen that the third-order ESO can estimate the system states and the real action of the unknown disturbance with high precision, even though in the presence of noise.

C. NPD Control and Disturbances Compensation

The linear proportional–integral–derivative (PID) controller is a widely used industrial controller which uses a linear combination of proportional, integral, and derivative action on the error to form the output of the controller. It is known that a linear combination of these components can achieve a compromised performance in terms of system response speed and stability. A nonlinear combination can provide an additional degree of freedom to achieve much improved system performance while still maintaining the advantages of simplicity and feasibility of linear PID control [30]–[33]. For this ADRC, an NPD control is used to synthesize the control action.

Once the high-quality differential signal has been obtained, the essential elements error e and its change c in classical PD control can be replaced by the following two components:

$$\begin{cases} e(k) = r_1(k) - z_1(k) \\ c(k) = r_2(k) - z_2(k). \end{cases} \quad (18)$$

Therefore, the primary control action of this ADRC can be synthesized [33]

$$u_0 = k_p \text{fal}(e, \alpha_3, \delta) + k_d \text{fal}(c, \alpha_4, \delta) \quad (19)$$

where $k_p, k_d, \alpha_3, \alpha_4$, and δ are design parameters.

With the feedforward control of the estimate of the real action of the unknown disturbances to cancel out their influences, the total control action of the proposed ADRC can be derived as

$$u = u_0 - \frac{z_3(k)}{b} \quad (20)$$

where b is a design parameter.

In fact, the above-described NPD control can be viewed as a magnitude-based gain-scheduling scheme. To further clarify this, we introduce the following equivalent gains, based on the

$$\begin{cases} \varepsilon_0 = z_1(k) - y(k) \\ z_1(k+1) = z_1(k) + h(z_2(k) - \beta_1 \varepsilon_0) \\ z_2(k+1) = z_2(k) + h(z_3(k) - \beta_2 \text{fal}(\varepsilon_0, \alpha_1, \delta_1)) \\ \vdots \\ z_n(k+1) = z_n(k) + h(z_{n+1}(k) - \beta_n \text{fal}(\varepsilon_0, \alpha_{n-1}, \delta_1) + b_0 u(k)) \\ z_{n+1}(k+1) = z_{n+1}(k) - h\beta_{n+1} \text{fal}(\varepsilon_0, \alpha_n, \delta_1) \end{cases} \quad (14)$$

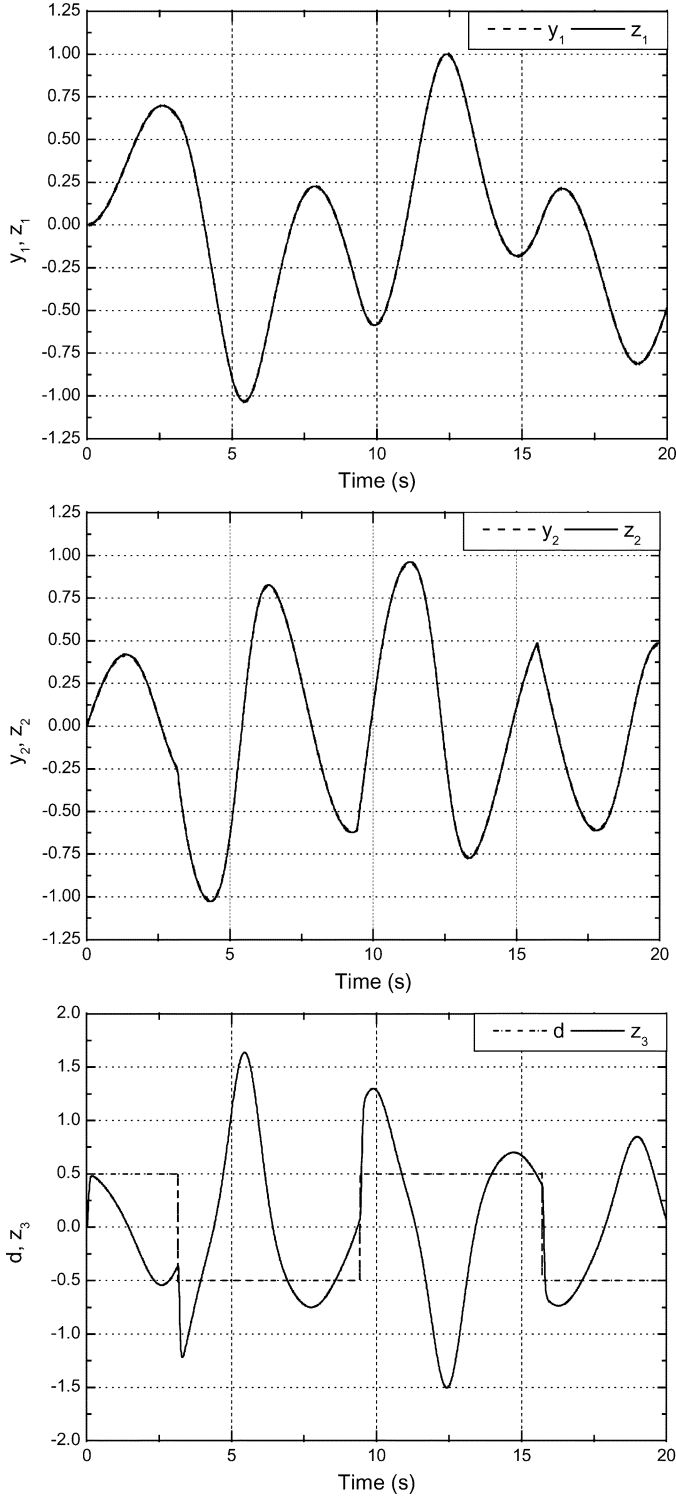


Fig. 4. Outputs of the third-order ESO and the system.

property of $\lim_{x \rightarrow 0} (\text{fal}(x, \alpha, \delta)) / (x) = \delta^{\alpha-1}$ of the nonlinear function $\text{fal}(x, \alpha, \delta)$:

$$\begin{cases} k_p^* = \beta_p \text{fal}(e, \alpha_3, \delta) / e \\ k_d^* = \beta_d \text{fal}(c, \alpha_4, \delta) / c. \end{cases} \quad (21)$$

It can be seen that the automatic gain adjustment is realized by the function $\text{fal}(x, \alpha, \delta)$. First of all, we can determine the variation of the function $\text{fal}(x, \alpha, \delta)$. A typical plot of the function

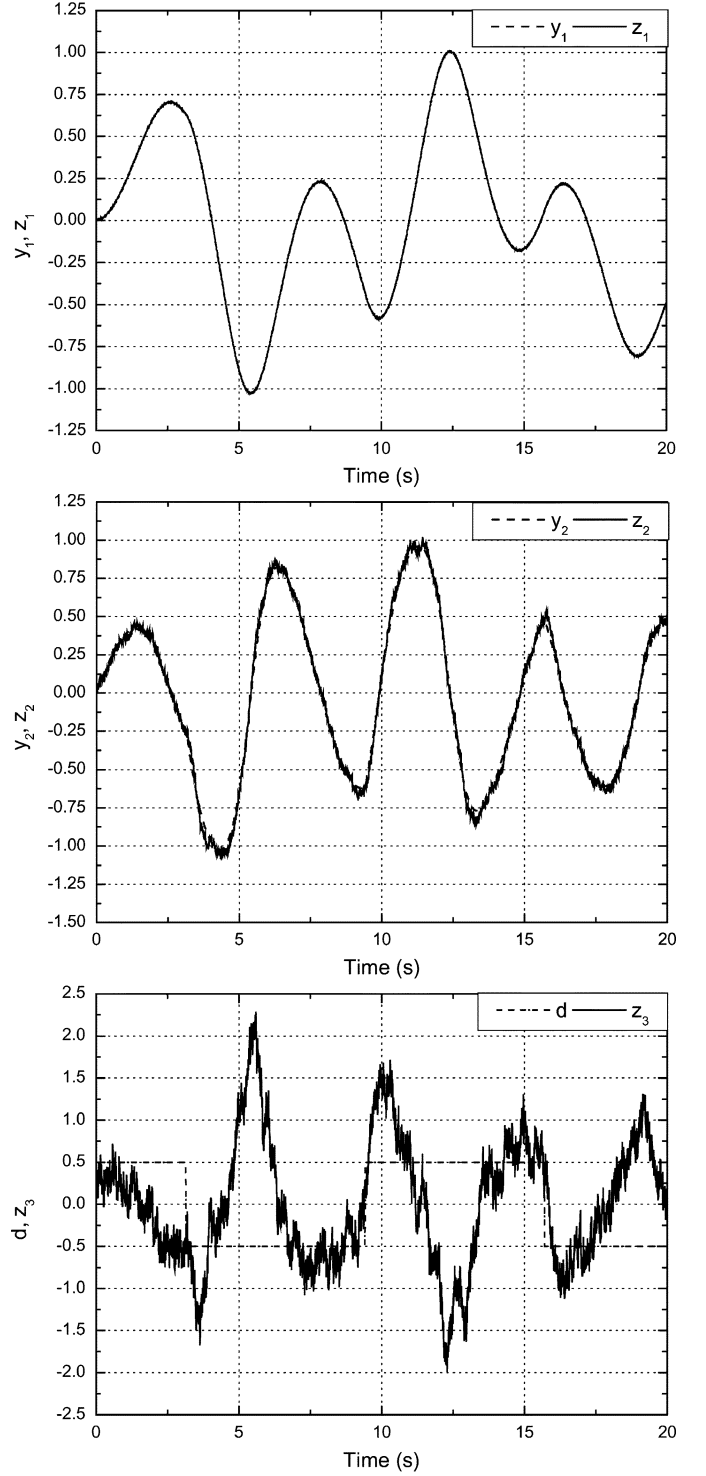


Fig. 5. Outputs of the third-order ESO and the system with noise.

$\text{fal}(x, \alpha, \delta)$ with $\alpha = 0.25, 1.0$, and 1.5 while $\delta = 0.01$ is illustrated in Fig. 6. The equivalent gains k_p^* and k_d^* versus the state are shown in Fig. 7, with $k_p = 1.0, k_d = 1.0, \delta = 0.01, \alpha_3 = 1.5$, and $\alpha_4 = 0.75$. It can be seen that such variation of the gains results in rapid transit with elaborate damping. When the error is large, the enlarged proportional gain can generate a large correction to drive the system output to its goal rapidly. As the error diminishes, the reduced proportional gain and the enlarged derivative gain can generate an elaborate small correction to

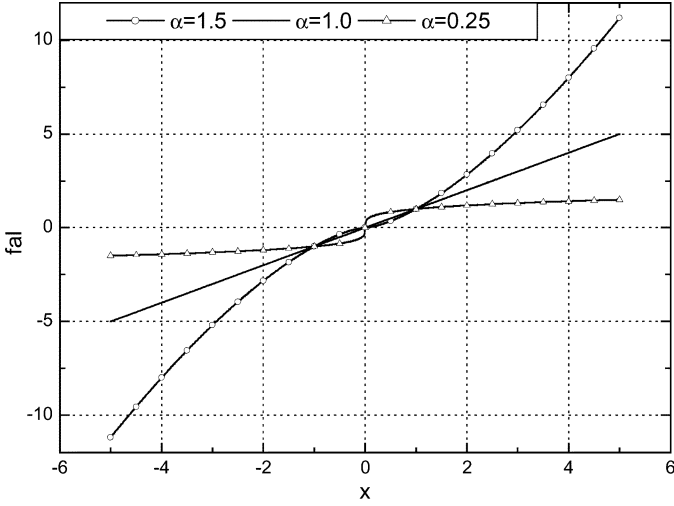
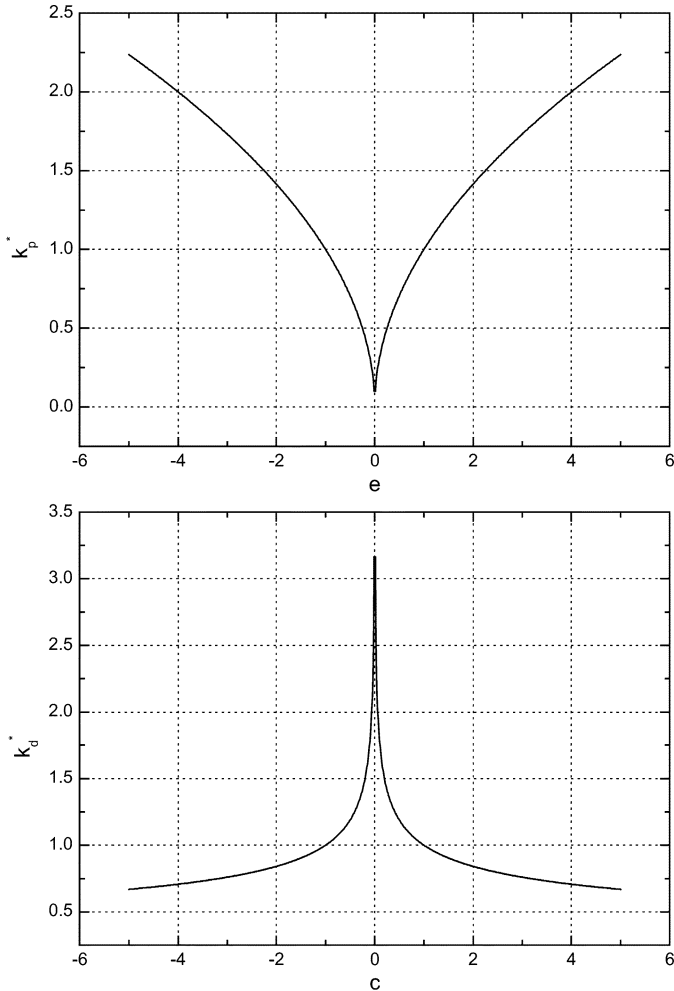
Fig. 6. $\text{fal}(x, \alpha, \delta)$ function.

Fig. 7. Variation of the equivalent gains of the NPD control.

prevent excessive oscillations and large overshoots. Because of this automatic gain adjustment, the NPD control used exhibits the advantage of high initial proportional gain and low derivative gain to obtain a fast response, followed by a low proportional gain and large derivative gain to prevent oscillations. To achieve these gain characteristics, the parameters of the function

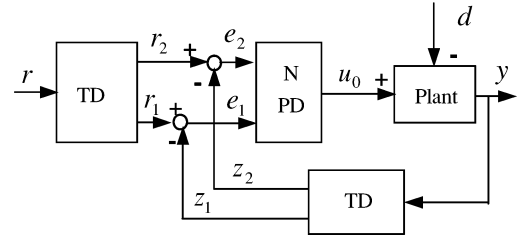


Fig. 8. Schematic diagram of NPD.

$\text{fal}(x, \alpha, \delta)$ should be determined as: $\delta = 0.001\text{--}0.1$, $\alpha_3 > 1.0$, and $0 < \alpha_4 < 1.0$.

For a linear PD control system, it has been proven that there exists a suitable choice of the PD controller parameters so that the overall closed-loop system is globally asymptotically stable for position control [34]–[37]. Referring to the NPD control used, the proportional and derivative gains k_p and k_d can be determined by using the following guidelines to retain the stability of the closed-loop system: 1) the appropriate equivalent gains k_p^* and k_d^* of a standard linear PD control are determined first, by using many available tuning rules for linear PD control and 2i) the proportional and derivative gains k_p and k_d of the NPD control must be smaller than k_p^* and k_d^* , due to the increased gains of the above nonlinear transformation, which can be determined by the nonlinear function $\text{fal}(x, \alpha, \delta)$.

IV. SIMULATIONS AND EXPERIMENTAL RESULTS

A. Simulations

The PMAC motor used in this research is an MSMA042A1G ac motor and the matched MSDA043A1A servo drive manufactured by Panasonic Inc. The specifications of the motor are: rated power, 0.4 kW; rated voltage, 200 V; rated current, 2.5 A; rated speed, 3000 r/min; $N = 4$, $R = 0.20 \, \Omega$, $L = 0.0012 \, \text{H}$; $K = 0.18 \, \text{N}\cdot\text{m}/\text{A}$; $J = 0.36 \times 10^{-4} \, \text{kg}\cdot\text{m}^2$; and $B = 8.56 \times 10^{-3} \, \text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$. An incremental optical encoder with a resolution of 4000 pulse/rev was used to provide the feedback position signal. The servo drive was set in the velocity mode, and the gains were selected as: $K_{P1} = 100$, $K_{I1} = 5$, $K_{P2} = 150$, and $K_{I2} = 80$, respectively.

The model equation given in (4) can be expanded in (22), under the assumption that the nonlinear friction is composed of the negative viscous friction and the Coulomb friction [11], [21]

$$J \frac{d\omega}{dt} = K \cdot N \cdot i_q - B \cdot \omega - f(\omega) \quad (22)$$

in which $f(\omega)$ can be expressed as [18], [21]

$$f(\omega) = \left(c_0 + c_1 e^{-c_2 |\omega|} \right) \text{sgn}(\omega) \quad (23)$$

where c_0 , c_1 and c_2 are the characteristic parameters, which can be determined by the experiment.

In fact, the friction compensation proposed in this paper is not based on any friction model; it is only used for the simulation. Combining the practical operating condition and the results presented in [21], the friction characteristic parameters were selected as follows: $c_0 = 2.142$, $c_1 = 2.922$, and $c_2 = 86.956$. Besides the above-determined parameters for the TD and ESO

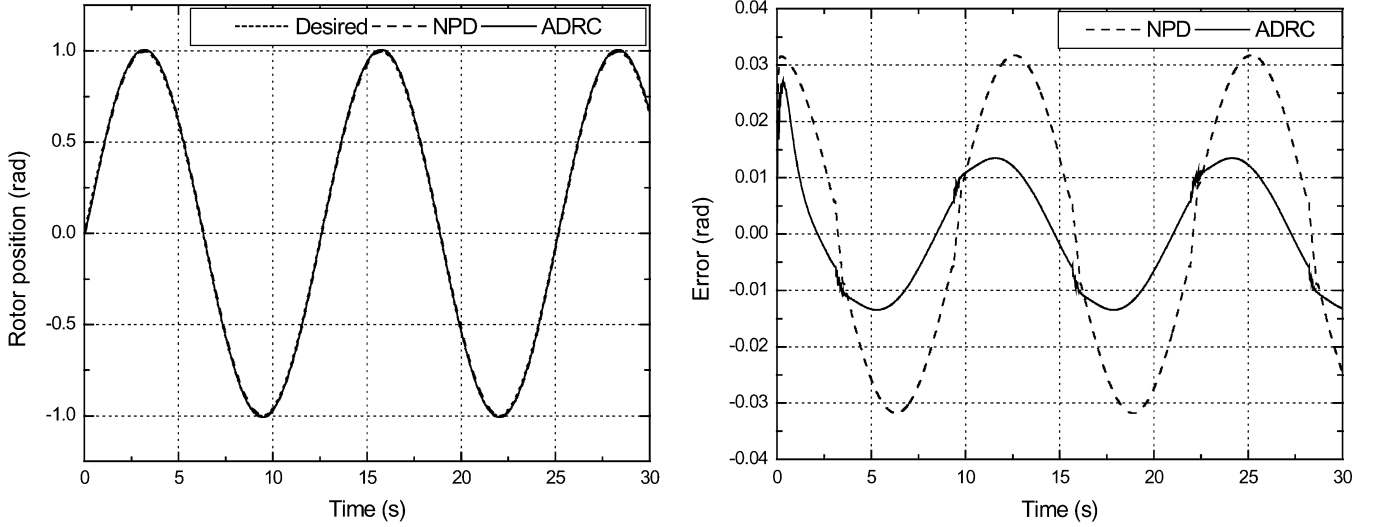


Fig. 9. Performance comparison between ADRC and NPD for $\theta^*(t) = \sin(0.5t)$ rad.

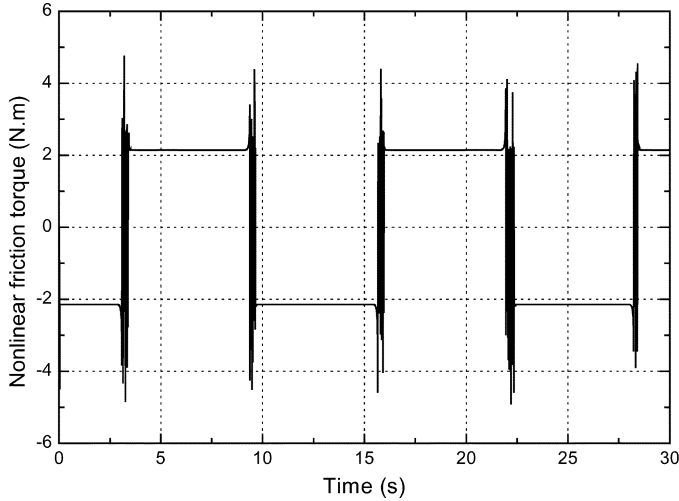


Fig. 10. Nonlinear friction torque for $\theta^*(t) = \sin(0.5t)$ rad.

in Section III, the other design parameters of the ADRC were determined as follows: $\delta = 0.01$, $\alpha_3 = 1.25$, $\alpha_4 = 0.5$, $k_p = 900$, $k_d = 3$, and $b = 1$.

For comparison, the NPD control expressed in (19) was also used for this motion control system. The schematic diagram of the NPD control is shown in Fig. 8. The proportional and derivative gains of the NPD control were determined as follows: $k_p = 1200$ and $k_d = 5$.

One way to access friction compensation is to measure the tracking error at relatively low speed, due to the fact that, at low speeds, a “stick-slip” phenomenon resulting from friction can lead to overshoot and large-amplitude limit cycles and will make the dominant contribution to the tracking error [2], [30]. Hence, the desired reference is a periodical sinusoidal profile of $\theta^*(t) = \sin(0.5t)$ rad. The simulation results are shown in Fig. 9 for the rotor position and tracking error. During the simulation, the nonlinear friction is shown in Fig. 10. It can be seen from these figures that the ADRC motion control PMAC motor system possesses better performance for friction compensation.



Fig. 11. Experimental setup.

B. Experimental Results

Fig. 11 shows the experimental setup, which consists of a six-channel D/A output card PCL-726 and a three-axes quadrature encoder card PCL-833. The control program was written in Borland C and run on an industrial computer made by Advantech Inc. The sampling period was determined as $T = 0.005$ s.

The following three experiments were carried out. The first experiment concerned the tracking performance of a periodical sinusoidal profile of $\theta^*(t) = \sin(0.5t)$ rad without load. The results with the ADRC and NPD are shown in Fig. 12. It can be seen that the ADRC achieves very good performance.

The second experiment concerned the performance robustness of the ADRC to parameter variation; a 6.0-kg payload was mounted on the motor shaft, so that the inertia load changed from $J = 0.36 \times 10^{-4}$ kg·m² to 5.05×10^{-4} kg·m². The experimental results with the ADRC and NPD to track $\theta^*(t) = \sin(0.5t)$ rad with this payload are shown in Fig. 13. Comparing Fig. 12 with Fig. 13, it shows that the ADRC achieves good tracking performance in spite of the change of inertia load.

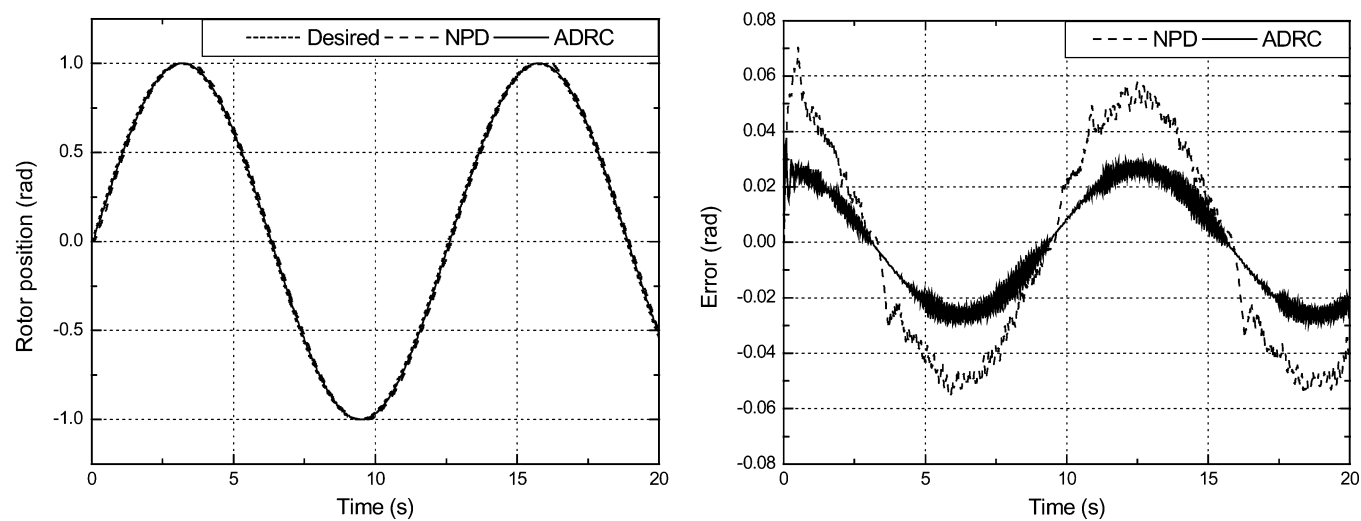


Fig. 12. Experimental comparison for $\theta^*(t) = \sin(0.5t)$ rad without load.

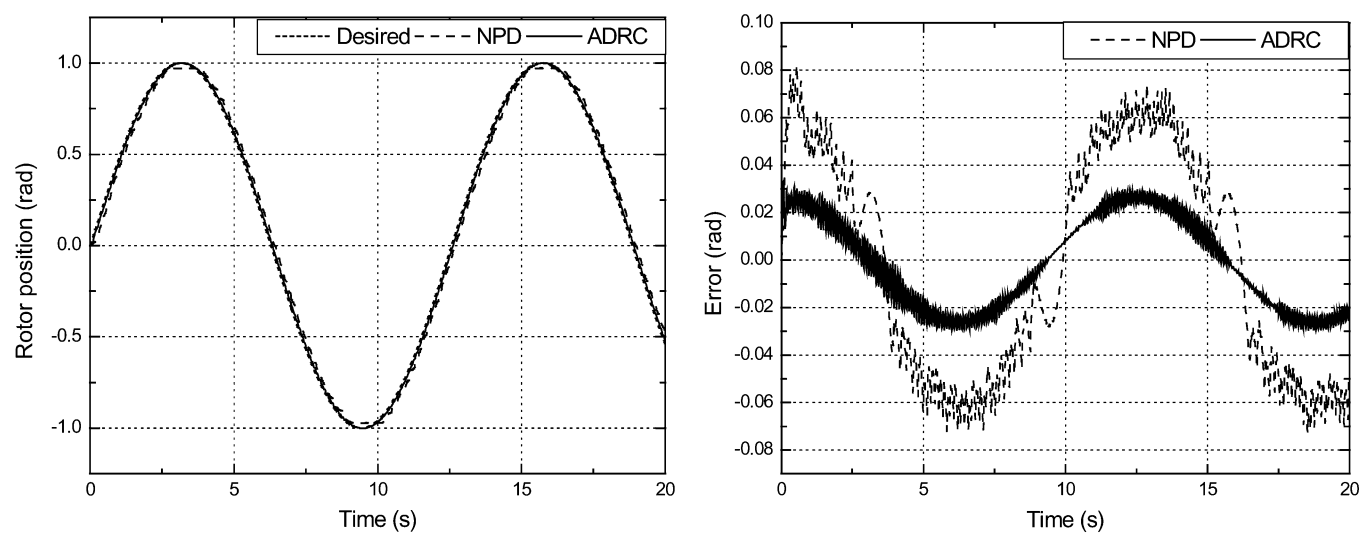


Fig. 13. Experimental comparison for $\theta^*(t) = \sin(0.5t)$ rad with load.

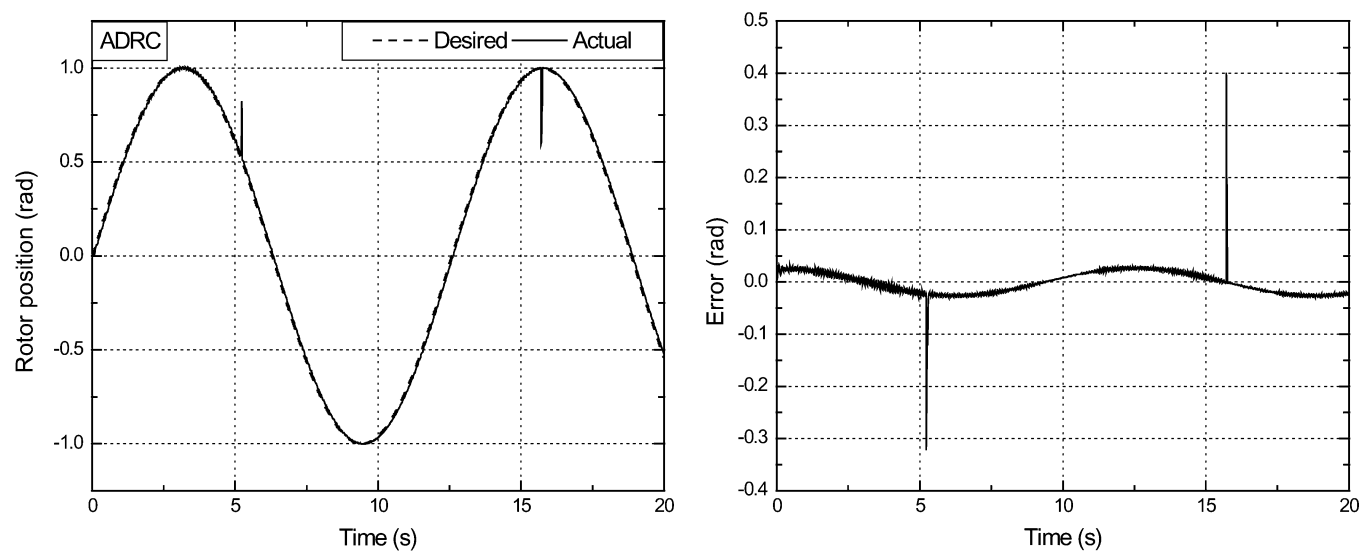


Fig. 14. Experimental result of ADRC with disturbance.

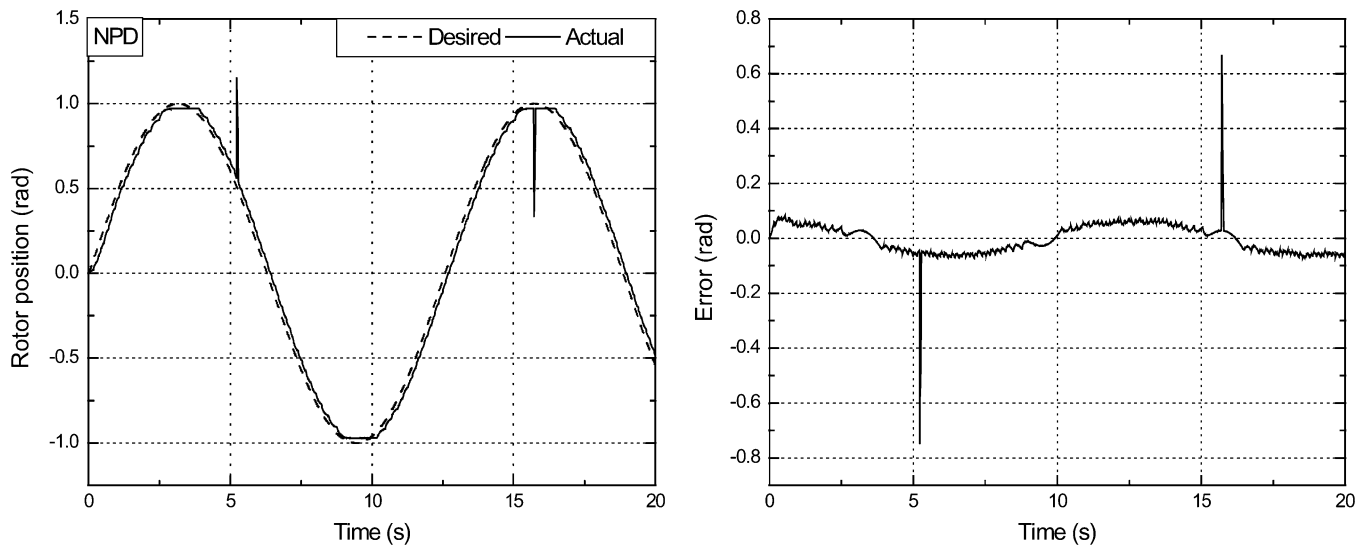


Fig. 15. Experimental result of NPD with disturbance.

The third experiment concerned the disturbance rejection of the proposed ADRC. A large step disturbance (a simulated position signal) described as follows is added:

$$d(t) = \begin{cases} 0, & \text{if } \cos(0.3t) > 0 \\ 0.5, & \text{if } \cos(0.3t) \leq 0. \end{cases} \quad (24)$$

The results obtained by ADRC and NPD are shown in Figs. 14 and 15, respectively. Comparing Fig. 14 with Fig. 15, it can be seen that the added large disturbance does not affect the performance of the ADRC much, except for the spike when the sudden change of the disturbance occurs. This result illustrates the disturbance rejection of the ADRC.

From these experiments, it can be summarized that the ADRC possesses a much higher robustness against parameters variation and disturbances and, therefore, has better performance.

V. CONCLUSION

An ADRC has been developed for high-performance robust motion control of PMAC motors. The proposed ADRC is designed directly in discrete time with a simple structure and fast computation. The proposed controller uses a TD in the feedforward path to solve the difficulties posed by low-order reference trajectories which are quantized at the sensor resolution and an ESO in the feedback path to estimate the unmeasured state and the real action of the unknown disturbances, including nonlinear friction. The NPD control is used to synthesize the control action. The resulting controller achieves high precision in the presence of both parametric uncertainties and bounded disturbances, which provides a simple way to implement the nonlinear friction compensation without any model. Comparative simulations and experimental results demonstrate the high performance of the proposed ADRC strategy.

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